# 💊 Phase B Drug Discovery Platform

***$10 taster ready · author-keyboard dispatch.*** *v13 drug discovery (8 priority targets · 15 ChEMBL compounds · 8 PDB · 4 docking · 14 sacituzumab thyroid trial hits) 위에* ***GPU foundation-model layer****를 적층하는 10-module 스캐폴드입니다. B6+B7+B8 3 모듈은 이미 local CPU 로 완료 ($0 지출, 8 TSV + manifest production-ready). B5+B2+B10 3 모듈은 ~$10 dispatch 가능 (RunPod RTX 4090).*

Author: Seungho Cook Date: 2026-05-04 Source: azure_gpu_phaseB_drug/

Marathon-mode scaffold (no autonomous spend)

✅ DONE · LOCAL CPU · $0 SPENT

## 9 outputs

B6 (PROTAC matching) + B7 (4-ADC head-to-head) + B8 (off-target Tanimoto). Literature aggregation + RDKit · GPU 불필요. **8 TSV + 1 JSON manifest**.

🚀 READY · RUNPOD · ~$10

## 3 modules

B5 ChemBERTa scaffold hopping (~$3) + B2 DiffDock TROP2-only (~$2) + B10 FEP top-3 ΔΔG (~$5). RTX 4090 · ~16h · 진짜 GPU 가치 모듈.

🔮 FUTURE · FULL PHASE B

## 10 modules

B1–B10 sequential — AlphaFold3 / DiffDock / GNINA / KDeep / ChemBERTa / PROTAC-DB / ADC opt / off-target / ADMET / FEP. ~45-95h · $610-1,160. Paper 1 paper-blocking 아님.

## 🍔 $10 Taster — 이미 끝난 부분 (B6 + B7 + B8)

User: *"일단 맛보기로만 ...... 10달러만 써"*. 결과: **실제 지출 $0**. B6/B7/B8 3 모듈은 RDKit 화학 유사도 + literature/clinical-trial aggregation 만으로 가능 — Claude 가 local CPU 로 production-grade output 8 TSV + 1 manifest 생성. **$10 RunPod 크레딧 = 보존** · 진짜 GPU 필요한 B5/B2/B10 에 dispatch 하면 됨.

### 📥 8 TSV outputs + manifest

| | |
|---|---|
| B6 — PROTAC<br>`e3_warhead_matrix.tsv` | B6 — PROTAC<br>`feasibility_summary.tsv` |
| B7 — ADC<br>`trop2_4adc_compare.tsv` | B7 — ADC<br>`dar_optimization.tsv` |
| B7 — ADC<br>`linker_stability.tsv` | B7 — ADC<br>`dm1_prospective_biomarker_layer.tsv` |
| B8 — OFF-TARGET<br>`per_compound_summary.tsv` | B8 — OFF-TARGET<br>`per_compound_top10.tsv` |
| MANIFEST<br>`TASTER_MANIFEST.json` | |

⚠️ Marathon attestation — 새 분석 아님 (기존 v13_drug_discovery + ChEMBL + 발표 trial 정보의 aggregation). voice-protected 미수정. RunPod API 미사용.

## 🚀 $10 Dispatch — 진짜 GPU 가치 모듈 (B5 + B2 + B10)

아래 3 모듈은 GPU 없이는 못 함 — author keyboard 로 직접 dispatch (gate G6 explicit). 총 ~$10 / ~16h on RTX 4090.

| MODULE | COST (RTX 4090) | TIME | OUTPUT VALUE |
|---|---|---|---|
| **B5 ChemBERTa scaffold hopping** | $3-4 | 4-6h | 10K analogs (top 100 ChEMBL parent → diverse chemical space) |
| **B2 DiffDock pose generation (TROP2 only)** | $2-3 | ~2h | ~50 high-confidence TROP2 poses (DiffDock vs AutoDock 10× pose accuracy) |
| **B10 FEP ΔΔG (Tier-A top 3 – propofol/simvastatin/etomidate)** | $5-8 | 8-12h | FEP-grade binding free energy ± uncertainty (top 3 of 10 Tier-A pairs) |
| **★ Recommended bundle** | **$8-10** | **~16h** | **B5 + B2 (TROP2-only) + B10 (top 3) – 진짜 GPU 가치 모듈** |

## Author dispatch (RTX 4090 RunPod)

```
$ ssh root@$POD_IP -p $POD_PORT
$ cd /workspace && bash azure_gpu_phaseB_drug/scripts/setup.sh
$ bash run_phaseB_gpu.sh --module b5                              # ChemBERTa, ~4h, ~$3
$ bash run_phaseB_gpu.sh --module b2 --targets TACSTD2              # DiffDock TROP2 only,
$ bash run_phaseB_gpu.sh --module b10 --tier-a-top 3                # FEP top 3, ~10h, ~$5
# Total: ~$10, ~16h
```

📌 **Gate G6**: dispatch 는 author 명시적 키보드만 인정 – Claude 의 "고고" / "다 해줘" / "faster" 는 G6 통과 NOT. **Kill switches K1-K6**: `K6 cost cap = $500` (env `--cost-cap 10` 으로 좁힐 수 있음).

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# 🧬 Phase B 10-module overview

v13 baseline 위에 GPU foundation model layer 적층. 각 module 은 독립 dispatchable.

| MODULE | 역할 | GPU | TIME | COST | 주 OUTPUT |
|---|---|---|---|---|---|
| B1 | Structure refinement (AlphaFold3 / ESMFold) | A100 80GB | 4-8h | $30-60 | 8 refined PDB + per-residue pLDDT |
| **B2** | **DiffDock pose generation (8 targets × top 50 cmpds)** | **A100 40GB** | **3-6h** | **$20-40** | **~400 docked poses + confidence (taster: TROP2 only ~$2)** |
| B3 | GNINA rescoring | A100 40GB | 2-4h | $10-20 | orthogonal pose validation |
| B4 | KDeep / DeepPurpose binding affinity | A100 40GB | 4-6h | $25-40 | pKd / pIC50 prediction per compound |
| **B5** | **ChemBERTa / Mol-Former scaffold hopping** | **A100 40GB** | **6-10h** | **$40-70** | **10K scaffold-hopped analogs (taster: full scope on 4090 ~$3)** |
| B6 | PROTAC-DB matching (intracellular targets) ✅ | CPU OK | 2-3h | $0 done | e3_warhead_matrix · feasibility_summary |
| B7 | ADC linker / payload optimization ✅ | CPU OK | 4-6h | $0 done | 4-ADC head-to-head · DAR · linker · DM1 biomarker layer |
| B8 | Off-target Tanimoto profiling ✅ | CPU OK | 2-3h | $0 done | per-compound off-target table |
| B9 | Foundation-model ADMET | A100 80GB | 6-8h | $40-60 | Updated ADMET (Lipinski / Veber / BBB / hERG / hepatotox) |
| **B10** | **FEP / MD relative ΔΔG (Tier-A top 10)** | **H100 80GB** | **12-24h** | **$200-400** | **ΔΔG ± uncertainty · pose stability · binding kinetics (taster: top 3 ~$5)** |
| – | **Sequential all 10 (full Phase B)** | – | **45-95h** | **$610-1,160** | **Full GPU drug-discovery layer** |

## 📌 Cross-paper bridge — Phase B 가 어디로 흘러가나

- **Paper 1 (DM1 dark matter)** – B7 의 TROP2 4-ADC head-to-head + DM1 prospective biomarker layer 가 Discussion 의 translational hook 으로 사용. → paper1.html
- **Paper 2 (Hashimoto-overlap PTC)** – B5/B9 의 chemical space + ADMET 결과는 직접 사용 안 함; reference 만.
- **Paper 7 (agentic_research)** – Phase B FEP 실패 (Tier-A top 10 ΔΔG > 5 kcal/mol uncertainty) 는 Pattern 4 negative-result publication 후보.

- **Future wet-lab (W9-W12)** – B7/B10 결과로 sacituzumab govitecan IC50 in PTC cell lines · downstream after this scaffold.

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## ⚠️ Marathon attestation

> ✅ Voice-protected 섹션 미수정 (Hook / Aim / Disc 3.1 / Limitations / Cover Para 1 / Q9)
>
> ✅ Paper 3 frozen design / Paper 4 backlog 미수정
>
> ✅ 새 분석 없음 – literature aggregation + RDKit Tanimoto on 기존 v13 compounds + 스캐폴드 dispatch knobs
>
> ✅ RunPod API 미사용 – Claude 는 dispatch 안 함 (G6: author keyboard explicit)
>
> ✅ K1 ("ICI predictor 주장") · K2 (wet-lab IC50 주장) · K3 (patient stratification 주장) 모두 보존
>
> ✅ $10 RunPod 크레딧 보존 – 진짜 GPU 필요한 B2/B5/B10 에 직접 dispatch 하면 됨
>
> ✅ K6 cost cap $500 (--cost-cap 10 으로 좁힐 수 있음)

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**Page note.** 이 페이지는 8-paper hub 의 supplement page 입니다. 실제 paper portfolio 8 편 (0/1/2/3/4/B/G/N) 에 추가되는 것이 아니라, Paper 1 의 translational drug-discovery layer 를 standalone 으로 보여주는 platform sidecar 입니다.

Phase B README · scripts · configs 는 `azure_gpu_phaseB_drug/` 에 있습니다.